THE ITERATION $x \mapsto a \bmod x$ HAS LENGTH $O_\varepsilon(a^{3/10+\varepsilon})$

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Abstract. For integers $a > b > 0$ let $b_1 = b$, $b_{i+1} = a \bmod b_i$, and let $P(a,b)$ be the number of positive terms of this sequence. Shallit asked whether $P(a,b) = O((\log a)^2)$. The bound $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$ of Erdős and Shallit was improved by Chase and Pandey to $a^{1/3-2/177+\varepsilon}$. We prove $P(a,b) \ll_\varepsilon a^{3/10+\varepsilon}$ by an elementary argument. The new ingredient is an exact identity for three consecutive steps of the iteration. It shows that a certain integer combination $D = dv - eu$ of the drops and quotient increments is always positive. It is also small whenever the three drops are small, and D together with the two increments confines the block to a curve with few integer points. The whole proof is formalised in Lean 4 with Mathlib, without `sorry`.

1. Introduction

Let $a > b > 0$ be integers. Put $b_1 = b$ and $b_{i+1} = a \bmod b_i$ while $b_i > 0$. The sequence is strictly decreasing until it reaches 0; if $b_{n+1} = 0$ we write $P(a,b) = n$. For example $P(35,22) = 7$, from $22, 13, 9, 8, 3, 2, 1, 0$. The number $P(a,b)$ is the length of the Pierce expansion of b/a [7, 8], since

$$\frac{b}{a} = \frac{1}{q_1} - \frac{1}{q_1 q_2} + \frac{1}{q_1 q_2 q_3} - \cdots, \qquad q_i = \lfloor a/b_i \rfloor.$$

Shallit [6] asked whether

$$P(a,b) = O\big((\log a)^2\big). \tag{1}$$

Shallit [8] proved $P(a,b) \ll a^{1/2}$ by "Archimedean" arguments. Erdős and Shallit [2] combined these with a divisor argument to get $P(a,b) \ll_\varepsilon a^{1/3+\varepsilon}$, and showed that $\max_b P(a,b)/\log a$ does not tend to 0. Chase and Pandey [1] went past the exponent $1/3$, proving $P(a,b) \ll_\varepsilon a^{1/3-2/177+\varepsilon}$ with exponential sum estimates (Bourgain's exponent pair), and showed that $\max_b P(a,b) \gg \log a/\log\log a$ for all large a. Shallit's talk pages [9] report a bound $n^{3/10}$, described as "LLM, unpublished so far". We have not seen that argument. The result below has the same exponent up to ε, and we make no claim about how the two arguments are related.

Theorem 1.1. *For every $\varepsilon > 0$ there is a constant C_ε such that*

$$P(a,b) \le C_\varepsilon\, a^{3/10+\varepsilon} \qquad \text{for all integers } 0 < b < a.$$

The proof is elementary. It uses only the division equations $a = q_i b_i + b_{i+1}$ and the divisor bound $\tau(n) \ll_\delta n^\delta$. As usual, the remainders are split into dyadic windows $[A, 2A)$, and we bound the number T of remainders in one window. There are two bounds:

$$T(T-1) \le 4A\,\Delta, \tag{2}$$

$$T^6 A^2 \le 995328\, a^3 \Delta \qquad (T \ge 12), \tag{3}$$

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where Δ is the largest value of the divisor function on $[1, 32a^7]$. Bound (2) is the Erdős–Shallit argument. Bound (3) is new. Multiplying (3) by the square of (2) eliminates A and gives $T^{10} \ll a^3\Delta^3$, which is the exponent $3/10$.

The new bound comes from three consecutive steps $x = b_i$, $y = b_{i+1}$, $z = b_{i+2}$, $w = b_{i+3}$. Let d, e, f be the drops, let u, v be the increments of the quotients, and put $D = dv - eu$. Then

$$D\,z = df + ude + (u-1)e^2. \tag{4}$$

So D is a *positive* integer, even though dv and eu largely cancel, and it is small when the drops are small. Also $(q, x) = (\lfloor a/b_i \rfloor, b_i)$ lies on the curve $xM(q) = aN(q) + D$, where M and N are explicit polynomials depending on (u, v) (Lemma 3.3). A resultant-type identity shows that each such curve has $a^{o(1)}$ admissible points (Lemma 4.1). The number of triples (u, v, D) is $\ll (B/T)^2 \cdot a/T^3$, where $B = a/A$, which gives (3).

Formal verification. Every step of the proof, including Theorem 1.1 in the form stated above, has been formalised in Lean 4 [3] using Mathlib [4]. Section 7 describes the formalisation. The constants in this paper are the constants in the Lean proof.

What is not proved. Theorem 1.1 is far from (1), and the method cannot reach it; see Section 8.

2. Preliminaries

2.1. **Notation.** Throughout, $a > b > 0$ are integers, $b_1 = b$, $b_{i+1} = a \bmod b_i$, and $P = P(a, b)$, so that $a > b_1 > b_2 > \cdots > b_P > 0 = b_{P+1}$. Write $q_i = \lfloor a/b_i \rfloor$, so

$$a = q_i b_i + b_{i+1}, \qquad 0 \le b_{i+1} < b_i \qquad (1 \le i \le P).$$

Let $\tau(n)$ be the number of positive divisors of n, and let

$$\Delta = \Delta(a) = \max_{1 \le n \le 32a^7} \tau(n).$$

Lemma 2.1 (Quotients increase)**.** *If $b_{i+1} > 0$ then $q_{i+1} \ge q_i + 1$.*

Proof. Since $b_{i+1} < b_i$ we have $(q_i + 1)b_{i+1} \le q_i b_i + b_{i+1} = a$, so $q_{i+1} = \lfloor a/b_{i+1} \rfloor \ge q_i + 1$. □

Lemma 2.2 (Divisor bound)**.** *For every integer $k \ge 1$ there is C_k with $\tau(n)^k \le C_k\, n$ for all $n \ge 1$.*

Proof. This is standard: $\tau(n)^k/n = \prod_{p^\alpha \| n} (\alpha+1)^k/p^\alpha$. Each factor with $p \ge 2^k$ is at most $(\alpha+1)^k/2^{k\alpha} \le 1$. Each of the fewer than 2^k factors with $p < 2^k$ is at most $\sup_\alpha (\alpha+1)^k/2^\alpha < \infty$. □

2.2. **Window runs.**

Definition 2.3. Let $A \ge 1$. A *window run* of length T is a sequence $x_0, x_1, \ldots, x_{T-1}$ of consecutive terms of the sequence (b_i), so $x_{j+1} = a \bmod x_j$, all lying in $[A, 2A)$.

In a window run, $x_0 > x_1 > \cdots > x_{T-1}$ and $x_j \le a$.

3. The curvature of three consecutive steps

Fix four consecutive terms

$$x = b_i, \quad y = b_{i+1} = a \bmod x, \quad z = b_{i+2} = a \bmod y, \quad w = b_{i+3} = a \bmod z,$$

with $x, y, z > 0$ (and $w \geq 0$). Write

$$q = \left\lfloor \frac{a}{x} \right\rfloor, \quad u = \left\lfloor \frac{a}{y} \right\rfloor - q, \quad v = \left\lfloor \frac{a}{z} \right\rfloor - \left\lfloor \frac{a}{y} \right\rfloor, \qquad d = x - y, \quad e = y - z, \quad f = z - w.$$

By Lemma 2.1, $u, v \geq 1$, and clearly $d, e, f \geq 1$. The division equations read

$$xq + y = a, \qquad y(q+u) + z = a, \qquad z(q+u+v) + w = a. \tag{5}$$

Definition 3.1. The *curvature* of the block is $D = dv - eu$.

Lemma 3.2 (Curvature identity)**.**

$$D\,z = df + ude + (u-1)e^2.$$

In particular $D \geq 1$.

Proof. Subtracting the first two equations of (5) gives $uy = qd + e$, hence $uz = uy - ue = qd + (1-u)e$. Subtracting the last two gives $vz = (q+u)e + f$. Therefore

$$(dv - eu)z = d\big((q+u)e + f\big) - e\big(qd + (1-u)e\big) = df + ude + (u-1)e^2.$$

The right-hand side is at least $df \geq 1$, and $z > 0$, so $D > 0$. □

Equivalently, the identity is the linear combination

$$Dz - \big(df + ude + (u-1)e^2\big) = d\,E_3 + (a - qx - x - y + z)\,E_2 + (-a + (q+u)y + y)\,E_1$$

of the three division equations E_1, E_2, E_3 of (5) (each written as left $-$ right $= 0$). This is how it is checked in Lean.

Lemma 3.3 (The fiber equation)**.** *Put* $M(q) = uq^2 + (u^2+u+v)q + v$ *and* $N(q) = uq + u^2 + v$. *Then*

$$x\,M(q) = a\,N(q) + D.$$

Proof. From (5), $d = (q+1)x - a$ and $e = y - z = (q+u+1)y - a = a(q+u) - q(q+u+1)x$. Substituting these into $D = dv - eu$ gives $D = x\big(uq(q+u+1) + v(q+1)\big) - a\big(u(q+u) + v\big)$, which is the claim. □

Lemma 3.4 (Size of the parameters)**.** *If* $x \leq a$ *then*

$$u\,yx \leq a(d+e), \qquad v\,zy \leq a(e+f), \qquad D\,z \leq (u+1)(d+e+f)^2.$$

Proof. From $uy = qd + e$ we get $uyx = (qx)d + ex \leq ad + ae$, since $qx \leq a$ and $x \leq a$. Similarly $vz = (q+u)e + f$ with $(q+u)y \leq a$ and $y \leq a$ gives the second bound. For the third, by Lemma 3.2, $Dz = df + ude + (u-1)e^2 \leq df + u\,e(d+e) \leq (1+u)(d+e+f)^2$. □

4. Few blocks share a triple (u, v, D)

Lemma 4.1 (Fiber count)**.** *Let* $a \geq 1$, $1 \leq u, v \leq a$ *and* $1 \leq D \leq 2a$. *Let* $M(q) = uq^2 + pq + v$ *with* $p = u^2 + u + v$, *and let* $L(q) = cq + \ell$ *with* $c = au$ *and* $\ell = a(u^2+v) + D$. *Then the set*

$$Q = \{q \geq 0 : \ xM(q) = L(q) \text{ for some integer } x \geq 1\}$$

has at most $\Delta = \max_{n \leq 32a^7} \tau(n)$ *elements.*

Proof. Put $U(q) = ucq + cp - u\ell$ and $R = u\ell^2 - cp\ell + c^2 v$. Expanding shows

$$c^2 M(q) - U(q)L(q) = R \tag{6}$$

identically in q. If $q \in Q$ with witness x, substituting $L(q) = xM(q)$ into (6) gives

$$R = M(q)\big(c^2 - U(q)x\big). \tag{7}$$

We bound the sizes first. From $u, v \le a$ and $D \le 2a$ we get $\ell \le 4a^3$, $p \le 3a^2$ and $c \le a^2$, so

$$|R| \le u\ell^2 + cp\ell + c^2 v \le 16a^7 + 12a^7 + a^5 \le 32a^7, \qquad c^2 \le a^4 \le 32a^7.$$

Case $R \neq 0$. By (7), $M(q)$ is a positive divisor of $|R|$. Since $u, p \ge 1$, the map $q \mapsto M(q)$ is strictly increasing on $q \ge 0$, hence injective. Therefore $|Q| \le \tau(|R|) \le \Delta$.

Case $R = 0$. Since $M(q) > 0$, (7) gives $U(q)x = c^2$. Here $c^2 > 0$ and $x \ge 1$, so $U(q)$ is a positive divisor of c^2. Since U is linear with positive leading coefficient uc, it is injective. Therefore $|Q| \le \tau(c^2) \le \Delta$. □

The case $R = 0$ cannot be dropped: it is the case in which M and L might share a factor.

5. Counting visits to a window

In this section $x_0, \dots, x_{T-1}$ is a window run in $[A, 2A)$, and $\Delta = \max_{n \le 32a^7} \tau(n)$.

Proposition 5.1 (Drop bound; after Erdős–Shallit)**.** *If $T \ge 2$ then $T(T-1) \le 4A\Delta$.*

Proof. Let $h_j = x_j - x_{j+1} \ge 1$ for $0 \le j \le T-2$. By telescoping, $\sum_j h_j = x_0 - x_{T-1} \le (2A-1) - A = A - 1$. Call j *good* if $h_j(T-1) \le 2A$ and *bad* otherwise. Each bad j has $h_j(T-1) \ge 2A+1$, so

$$|\,\mathrm{Bad}\,|\,(2A+1) \le (T-1)\textstyle\sum_j h_j \le (T-1)(A-1).$$

If $2|\,\mathrm{Bad}\,| \ge T-1$ this would give $(T-1)(2A+1) \le 2(T-1)(A-1)$, which is false. Hence $2|\,\mathrm{Bad}\,| < T-1$, and since $|\,\mathrm{Good}\,| + |\,\mathrm{Bad}\,| = T-1$ we get $2|\,\mathrm{Good}\,| \ge T$.

For good j we have $1 \le h_j \le H := \lfloor 2A/(T-1) \rfloor$. Also $a + h_j = q_j x_j + x_{j+1} + x_j - x_{j+1} = (q_j+1)x_j$, so $x_j \mid a + h_j$. Moreover $a + h_j \le 2a \le 32a^7$. Since $j \mapsto x_j$ is injective, at most $\tau(a+h) \le \Delta$ good indices have $h_j = h$. Hence $|\,\mathrm{Good}\,| \le H\Delta$, and

$$T(T-1) \le 2|\,\mathrm{Good}\,|(T-1) \le 2\Delta\, H(T-1) \le 4A\Delta.$$ □

Proposition 5.2 (Curvature bound)**.** *If $T \ge 12$ then $T^6 A^2 \le 995328\, a^3 \Delta$.*

Proof. For $0 \le j \le T-4$ let $s_j = x_j - x_{j+3}$ be the total drop of the block $x_j, x_{j+1}, x_{j+2}, x_{j+3}$. By telescoping,

$$\sum_{j=0}^{T-4} s_j = (x_0 + x_1 + x_2) - (x_{T-3} + x_{T-2} + x_{T-1}) \le 3(2A-1) - 3A = 3A - 3.$$

Call j *good* if $s_j T \le 12A$ and *bad* otherwise. Then $|\,\mathrm{Bad}\,|(12A+1) \le T(3A-3)$. If $4|\,\mathrm{Bad}\,| \ge T$ this would give $T(12A+1) \le 4T(3A-3)$, which is false. So $4|\,\mathrm{Bad}\,| \le T-1$. Since $|\,\mathrm{Good}\,| = T - 3 - |\,\mathrm{Bad}\,|$ and $T \ge 12$,

$$2|\,\mathrm{Good}\,| \ge 2T - 6 - \tfrac{T-1}{2} \ge T.$$

Fix a good j. Apply Section 3 to $x = x_j$, so that $y, z, w = x_{j+1}, x_{j+2}, x_{j+3}$, all in $[A, 2A)$. Write $s = s_j = d + e + f$. Lemma 3.4 gives $uA^2 \le uyx \le a(d+e) \le as$. Multiplying by T and using $sT \le 12A$ gives $uA^2 T \le 12aA$, that is,

$$u\,AT \le 12a, \qquad \text{and likewise} \qquad v\,AT \le 12a.$$

Also $DA \le Dz \le (u+1)s^2 \le 2us^2$. Multiplying by T^2 gives $DAT^2 \le 2u(sT)^2 \le 288uA^2$, so $DT^2 \le 288uA$, and then

$$D\,T^3 \le 288\,u\,AT \le 3456\,a.$$

Thus the triple (u, v, D) of a good block lies in $[1, U]^2 \times [1, D^*]$, where

$$U = \lfloor 12a/(AT) \rfloor \le a, \qquad D^* = \lfloor 3456a/T^3 \rfloor \le 2a.$$

The inequalities $U \le a$ and $D^* \le 2a$ hold because $AT \ge 12$ and $T^3 \ge 1728$.

Now fix such a triple and consider the good blocks that have it. By Lemma 3.3, each one gives a point $q = q_j = \lfloor a/x_j \rfloor$ of the set Q of Lemma 4.1. The map $j \mapsto q_j$ is injective on these blocks: q_j determines $x_j = L(q_j)/M(q_j)$, and x_j determines j. By Lemma 4.1, at most Δ good blocks share a triple. So $|\,\mathrm{Good}\,| \le U^2 D^* \Delta$, and

$$T^6A^2 = T\,(AT)^2\,T^3 \le 2\Delta\,(U\,AT)^2\,(D^*T^3) \le 2\Delta(12a)^2(3456a) = 995328\,a^3\Delta. \qquad \square$$

Corollary 5.3 (Window bound)**.** *If $T \ge 12$ then $T^{10} \le 63700992\,a^3\Delta^3$.*

Proof. Using $T \le 2(T-1)$ and the two propositions,

$$T^{10}A^2 \le 4\,(T^6A^2)\,\big(T(T-1)\big)^2 \le 4 \cdot 995328\,a^3\Delta \cdot (4A\Delta)^2 = 63700992\,a^3\Delta^3A^2.$$

Divide by A^2. $\square$

Remark 5.4. In terms of $B = a/A$, Propositions 5.1 and 5.2 say $T \ll a^{1/2+o(1)}B^{-1/2}$ and $T \ll a^{1/6+o(1)}B^{1/3}$. The first is better when $B > a^{2/5}$, and both equal $a^{3/10+o(1)}$ at $B = a^{2/5}$. Corollary 5.3 takes the product of the two bounds, so no case split is needed.

6. Proof of Theorem 1.1

Let $0 < b < a$. For $0 \le k \le \log_2 a$ let T_k be the number of indices $i \le P$ with $2^k \le b_i < 2^{k+1}$. Every b_i with $i \le P$ lies in $[1, a)$, so $P = \sum_{0 \le k \le \log_2 a} T_k$. Since (b_i) is decreasing, the indices counted by T_k are consecutive. So when $T_k > 0$ they form a window run in $[A, 2A)$ with $A = 2^k$.

Let $\varepsilon > 0$. Choose an integer $m > 7/(5\varepsilon)$, and let C_{3m} be the constant of Lemma 2.2 with $k = 3m$. The maximum Δ is attained at some $n_0 \le 32a^7$, so $\Delta^{3m} \le C_{3m} \cdot 32a^7$. If $T_k \ge 12$, Corollary 5.3 gives

$$T_k^{10m} \le 63700992^m\,a^{3m}\,\Delta^{3m} \le K\,a^{3m+7}, \qquad K = 63700992^m \cdot C_{3m} \cdot 32,$$

and hence

$$T_k \le K^{1/(10m)}\,a^{3/10+7/(10m)} \le K^{1/(10m)}\,a^{3/10+\varepsilon/2}.$$

If $T_k \le 11$ this bound holds with 11 in place of $K^{1/(10m)}$. Finally $\lfloor \log_2 a \rfloor + 1 \le C'_\varepsilon a^{\varepsilon/2}$. Summing over k,

$$P(a,b) \le (\lfloor \log_2 a \rfloor + 1)\big(11 + K^{1/(10m)}\big)a^{3/10+\varepsilon/2} \le C_\varepsilon\,a^{3/10+\varepsilon}. \qquad \square$$

7. The Lean formalisation

The repository accompanying this paper contains a Lean 4 project (toolchain `v4.35.0`, Mathlib `v4.35.0`) that proves Theorem 1.1. The final statement is

```
theorem P_le : ∀ ε > (0 : ℝ), ∃ C > 0, ∀ a b : ℕ, 0 < b → b < a →
    (P a b : ℝ) ≤ C * (a : ℝ) ^ ((3 : ℝ) / 10 + ε)
```

where `P a b` is defined by well-founded recursion as the number of positive terms ($P(a,0) = 0$, $P(a,b) = P(a, a \bmod b) + 1$ for $b > 0$), and the file checks $P(35, 22) = 7$. The command `#print axioms P_le` reports only `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no `native_decide`.

Paper	Lean	File
Definition of $P(a,b)$	`P`, `orb`	`Defs.lean`
Lemma 2.1	`div_lt_div_mod`	`Block.lean`
Lemma 3.2	`curvature_identity`, `one_le_bD`	`Block.lean`
Lemma 3.3	`fiber_identity`, `block_fiber_eq`	`Block.lean`
Lemma 3.4	`bu_mul_le`, `bv_mul_le`, `bD_mul_le`	`Block.lean`
Lemma 4.1	`fiber_poly_identity`, `fiber_card_le`	`Fiber.lean`
Window runs	`Window`, `DivBound`	`Window.lean`
Proposition 5.1	`drop_bound`	`Window.lean`
Proposition 5.2	`curvature_bound`	`Window.lean`
Corollary 5.3	`window_bound`	`Window.lean`
Dyadic decomposition	`P_eq_sum_visits`, `visits_window`	`Main.lean`
Theorem 1.1	`P_le`	`Main.lean`
Lemma 2.2	`exists_divisors_power_bound`	`DivisorPowerBound.lean`

Some details of the formalisation:

- The algebraic identities (Lemmas 3.2, 3.3 and (6)) are proved over $\mathbb{Z}$ by the tactic `linear_combination`, using the explicit coefficients above. They are then transferred to $\mathbb{N}$ by writing $z = w + f$, $y = z + e$, $x = y + d$, which avoids truncated subtraction.
- The curvature is the natural number $\texttt{bD}\,a\,x = (x-y)v - (y-z)u$. Lean first proves $eu < dv$, so this truncated subtraction is the true value.
- The counting arguments use `Finset.card_le_mul_card_image_of_maps_to`: the number of good indices is at most the size of the parameter box times the largest fiber.
- The divisor bound $\tau(n)^k \le C_k n$ (Lemma 2.2) is the file `DivisorPowerBound.lean` from the `openai/math` repository [5] (Apache-2.0), vendored unchanged except for its import lines.
- The project has about 1000 lines of Lean. `lake build` rebuilds it in under a minute once Mathlib is built.

8. Limits of the method

The argument counts the triples (u, v, D) independently over their ranges. Even if the divisor loss in Lemma 4.1 were replaced by an absolute constant, the bound would still be $T \ll (B/T)^2\, a/T^3$. Combined with $T \ll A^{1/2}$, this still gives the exponent $3/10$. So removing the ε, sharpening constants or improving the divisor count cannot give a logarithmic bound. Further progress needs to restrict *which* triples can occur together along a single orbit.

There is no obvious monotonicity to exploit. The curvatures D_i of successive blocks need not decrease. For $a = 13242$ the orbit of 255 begins

$$255,\ 237,\ 207,\ 201,\ 177,\ 144,\ 138,\ 132,\ \ldots$$

with quotients $51, 55, 63, 65, 74, 91, 95, 100, \ldots$. The curvatures of the first five blocks are $24, 12, 6, 111, 30$, all for blocks that lie within a factor of two.

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References

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